

Mohamad Dikshie Fauzie

PRINCIPAL INFRASTRUCTURE ARCHITECT · LOW-LATENCY & DISTRIBUTED SYSTEMS

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Summary

Principal Infrastructure Architect and Systems Engineer with 10+ years of mission-critical production experience co-managing low-latency infrastructure at Japan's largest alternative trading venue (PTS), backed by a Ph.D. from Keio University (supervised by Internet Hall of Fame pioneer Prof. Jun Murai and quantum networking pioneer Prof. Rodney Van Meter) and an ITB Physics degree. Proven record delivering zero-downtime datacenter migrations, 50% rack power balancing optimizations, 50% storage throughput gains via Dell PowerScale (Isilon), and end-to-end telemetry automation. Combines deep physical-layer discipline (power/cooling capacity planning, optical metrology) with advanced systems engineering: sub-nanosecond precision timing (NPL Time & Frequency), low-latency TCP/UDP kernel tuning, NetFPGA, AI cluster infrastructure (NVIDIA certified), and cloud-native virtualization.

Skills

Systems & OS	Low-Latency Architectures, High-Performance Systems (HPC), AI/GPU Clusters, Linux (RHEL, Ubuntu), FreeBSD, Kernel Tuning
Networking	TCP/UDP, BGP, OSPF, IPv6 Core, IEEE 1588v2 PTP, Sub-Nanosecond Timing, SDN, P2P/Mesh Topologies, CDN, NetFPGA
Storage & Cloud	Dell PowerScale (Isilon) Scale-Out Storage, Kubernetes, Docker, OpenStack, ONAP, OPNFV, AWS Cloud Fundamentals
Facilities & DC	Rack PDU Power Telemetry & Balancing, Facility Thermal/Capacity Planning, Server Migration, Optical Metrology (SFPs/QSFPs)
Automation	Python, Bash/Shell, Git, SNMP Telemetry, Open-Source Monitoring & Inventory Systems, LaTeX
Languages	English (Fluent / Professional), Japanese (Working Proficiency), Indonesian (Native)

Work Experience

Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

DATA CENTER ENGINEER (CO-MANAGER) / INFRASTRUCTURE LEAD

Oct. 2014 - Present

- Co-managed 24/7 mission-critical, ultra-low-latency datacenter operations for Japan's primary alternative equity trading venue, maintaining 99.999% infrastructure availability under FSA regulatory compliance.
- Led end-to-end datacenter server migration for power-constrained production systems; orchestrated risk mitigation protocols and executed with minimal downtime, delivering 2 weeks ahead of schedule and unlocking 50% expansion capacity.
- Spearheaded rack-level PDU power telemetry across high-density server racks, achieving a 50% improvement in workload power balancing and optimizing facility power and cooling capacity planning.
- Architected and deployed enterprise scale-out storage on Dell PowerScale (Isilon) across cross-functional teams, achieving a 50% increase in server throughput over legacy storage and maximizing shared storage efficiency.
- Designed and deployed an online inventory tracking system for physical infrastructure hardware and optical consumables (SFPs/QSFPs, patch panels, cabling), drastically accelerating failure discovery, MTTR, and procurement.
- Partnered with Unix infrastructure engineering to integrate an open-source hardware discovery platform with automated system monitoring, streamlining asset discovery, fault detection, and capacity planning.
- Automated executive monthly capacity, power, and SLA reporting using open-source telemetry pipelines, eliminating manual reporting overhead and boosting operational transparency.
- Directed datacenter room buildouts and server rack installations for critical internal trading systems and client co-location racks, achieving full operational readiness one week ahead of schedule.

Keio University - JICA JST Indo Project

Tokyo, Japan

RESEARCH ASSISTANT (DISTRIBUTED SYSTEMS & NETWORKS)

Apr. 2011 - Sep. 2014

- Researched distributed network topologies, energy efficiency, and power consumption in large-scale peer-to-peer (P2P) networks and Content Delivery Networks (CDNs).
- Applied game theory models to information dissemination using probabilistic gossip protocols, analyzing graph dynamics in dynamic swarms.
- Published peer-reviewed research papers in international IEEE and IEICE journals and conference proceedings.

Keio University

Tokyo, Japan

SYSTEM ADMINISTRATOR (CAMPUS CORE INFRASTRUCTURE)

Oct. 2007 - Sep. 2014

- Administered campus server and core network infrastructure (DNS, email, web servers, BGP/OSPF routers, and switches) in coordination with lab managers, unifying administrative workflows for high operational reliability.

Certifications & Specialized Training

2024	AI Infrastructure and Operations Fundamentals , NVIDIA (Deep Learning Institute)	Online
2024	Foundations of Project Management , Google Career Certificates	Online
2022	Introduction to Time and Frequency Measurement , National Physical Laboratory (NPL), UK	Teddington, UK
2020	Introduction to Metrology (Measurement Standards) , National Physical Laboratory (NPL), UK	Teddington, UK
2010	Introduction to NetFPGA (High-Speed Packet Processing) , Stanford University	Tokyo, Japan
2023	AWS Certified Cloud Practitioner , Amazon Web Services (AWS)	Online
2022 - 2023	Machine Learning & Mathematics for Machine Learning , Stanford University / DeepLearning.AI	Online
2019	Cloud Infrastructure, Kubernetes & OpenStack , The Linux Foundation	Online
2019	Network Transformation: 5G, Core Network, Cloud & NFV , Intel	Online
2019	Carrier Network Automation: ONAP, OPNFV & Open Source Networking , The Linux Foundation	Online
2022	Financial Risk Management & Capital Markets , New York Institute of Finance / Interactive Brokers	Online

Publications

Energy Consumption in Peer Assisted CDN	<i>International Journal of Advanced Research in Computer Science (IJARCS) Vol.4 No.9</i> Aug. 2013
LEAD AUTHOR	
• Modeled and evaluated energy efficiency of peer-assisted edge CDN architectures versus traditional centralized distribution.	
Assessing The Dynamics of BitTorrent Swarm Topologies Using The Peer Exchange Protocol	<i>IEICE Transactions on Communications (Special Issue on Information Network Science, Vol.E95-B, No.05)</i> 2012
LEAD AUTHOR	
• Empirically inferred dynamic overlay topologies and peer discovery mechanisms using the Peer Exchange (PEX) protocol.	
A Temporal View of The Topology of Dynamic BitTorrent Swarms	<i>IEEE INFOCOM Workshop on Network Science for Communication Networks (DOI: 10.1109/INFCOMW.2011.5928939)</i> 2011
LEAD AUTHOR	
• Analyzed temporal graph evolution and topological robustness of large-scale distributed swarms under dynamic churn.	

Education

Keio University	<i>Tokyo, Japan</i>
PH.D. IN MEDIA AND GOVERNANCE (INTERNET ARCHITECTURE & DISTRIBUTED SYSTEMS)	<i>Graduated Oct. 2014</i>
• Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).	
• Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer), Dr. Kenjiro Cho, Prof. Hiroshi Kusumoto.	
Institut Teknologi Bandung (ITB)	<i>Bandung, Indonesia</i>
M.T. (MASTER OF TECHNOLOGY IN ELECTRICAL ENGINEERING)	<i>Graduated Oct. 2004</i>
• Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.	
Institut Teknologi Bandung (ITB)	<i>Bandung, Indonesia</i>
S.Si. (BACHELOR OF SCIENCE IN PHYSICS)	<i>Graduated Oct. 2000</i>
• Researched data assimilation methods in 1D hydrodynamic fluid systems (numerical modeling, wave equations, fluid mechanics).	